

Kochen–Specker uncolorability in cyclotomic fields requires exactly $6 \mid n$

Michael Kernaghan¹

¹*Pacific Quantum Systems, Vancouver, Canada*

(Dated: February 18, 2026)

We prove that the ray set generated by the n th roots of unity in $\mathbb{C}^3$ is Kochen–Specker uncolorable if and only if $6 \mid n$. Sufficiency follows from the embedding of Cabello’s Eisenstein 33-vector KS set. Necessity is proved by algebraic analysis of vanishing sums among roots of unity: when n is odd and $3 \mid n$, a projective collapse argument shows every triad is isolated (no two share a ray), so the coloring problem decomposes; when $3 \nmid n$ and $2 \mid n$, the orthogonality graph on each coordinate plane is a perfect matching, admitting an explicit coloring. The factorization $6 = 2 \times 3$ mirrors the result of Cortez, Morales, and Reyes that $\mathbb{Z}[1/6]$ is the minimal ring supporting KS contextuality in dimension 3: the arithmetic conditions $2 \mid n$ (two-term cancellation) and $3 \mid n$ (three-term phase cancellation) are independently necessary and jointly sufficient.

INTRODUCTION

The Kochen–Specker (KS) theorem [?] establishes that quantum measurements in dimension $d \geq 3$ cannot be explained by noncontextual hidden-variable theories, by exhibiting a finite set of rays admitting no consistent $\{0, 1\}$ -valued coloring. A natural source of KS constructions is the cyclotomic family: for each positive integer n , the alphabet $\{0\} \cup \{\zeta^k : k = 0, \dots, n-1\}$ with $\zeta = e^{2\pi i/n}$ generates a ray set S_n in $\mathbb{C}^3$ via all projectively distinct nonzero vectors with coordinates drawn from this alphabet.

Cabello [?] showed that S_6 contains a 33-vector KS set built from the Eisenstein integers $\mathbb{Z}[\omega]$ ($\omega = e^{2\pi i/3}$), which he identified as the “simplest” KS set by the criterion of fewest bases. Cortez, Morales, and Reyes [?] proved that $\mathbb{Z}[1/6]$ is the minimal ring extension of $\mathbb{Z}$ exhibiting KS contextuality in dimension 3—that is, the primes 2 and 3 are both necessary.

These results suggest a sharp characterization: KS-uncolorability in the cyclotomic family should require exactly $6 \mid n$. We prove this.

The ray set S_n generated by the n th roots of unity in $\mathbb{C}^3$ is KS-uncolorable if and only if $6 \mid n$.

PRELIMINARIES

A *ray* in $\mathbb{C}^3$ is a one-dimensional subspace. Two rays v, w are *orthogonal* if $\langle v|w \rangle = \sum_k \bar{v}_k w_k = 0$. An *orthogonal triad* is a set of three mutually orthogonal rays. A *KS coloring* assigns each ray a value in $\{0, 1\}$ such that orthogonal rays cannot both receive 1 and every triad contains exactly one 1. A ray set is *KS-uncolorable* if no such coloring exists.

For $\zeta = e^{2\pi i/n}$, the ray set S_n consists of all projectively distinct nonzero vectors (a_1, a_2, a_3) with each $a_k \in \{0\} \cup \{\zeta^j : 0 \leq j < n\}$. The orthogonality condition $\langle v|w \rangle = 0$ reduces to a vanishing sum of terms, each of

the form ζ^m or 0.

PROOF OF THEOREM ??

Sufficiency

If $6 \mid n$, then $\zeta^{n/6}$ generates the 6th roots of unity as a subgroup of $\langle \zeta \rangle$, so the alphabet for $n = 6$ embeds into that for general n . Since S_6 contains Cabello’s 33-vector KS set [?], S_n is KS-uncolorable.

Necessity: vanishing-sum analysis

We show that if $6 \nmid n$, then S_n admits a valid KS coloring. The proof rests on two lemmas characterizing when vanishing sums of roots of unity exist.

[Three-term vanishing sums] The equation $\zeta^a + \zeta^b + \zeta^c = 0$ has a solution with $\zeta = e^{2\pi i/n}$ if and only if $3 \mid n$.

Dividing by ζ^a : $1 + \zeta^p + \zeta^q = 0$ where $p = b - a$, $q = c - a$. Setting $\alpha = \zeta^p$, we need $\zeta^q = -1 - \alpha$ with $|\zeta^q| = 1$. Computing $|-1 - \alpha|^2 = 2 + 2\cos(2\pi p/n)$, the condition $|-1 - \alpha| = 1$ forces $\cos(2\pi p/n) = -1/2$, hence $p/n \in \{1/3, 2/3\}$, which requires $3 \mid n$. Conversely, if $3 \mid n$, then $1 + \zeta^{n/3} + \zeta^{2n/3} = 0$.

[Two-term vanishing sums] The equation $\zeta^a + \zeta^b = 0$ has a solution if and only if $2 \mid n$.

$\zeta^a + \zeta^b = 0$ iff $\zeta^{b-a} = -1$ iff $-1 \in \langle \zeta \rangle$ iff $2 \mid n$.

These lemmas constrain which orthogonalities can occur in S_n . Two all-nonzero rays $v = (\zeta^{a_1}, \zeta^{a_2}, \zeta^{a_3})$ and $w = (\zeta^{b_1}, \zeta^{b_2}, \zeta^{b_3})$ are orthogonal iff $\zeta^{b_1-a_1} + \zeta^{b_2-a_2} + \zeta^{b_3-a_3} = 0$ —a three-term vanishing sum. By Lemma ??, this requires $3 \mid n$. Similarly, orthogonality between one-zero rays in the same coordinate plane requires a two-term vanishing sum, hence $2 \mid n$ by Lemma ??.

We now treat the three cases where $6 \nmid n$.

Case 1: $3 \nmid n, 2 \nmid n$

No vanishing sums of any length exist among the n th roots. The only orthogonalities are axis-complementary (e.g., $(\zeta^a, 0, 0) \perp (0, \zeta^b, \zeta^c)$). The sole triad is $\{[1:0:0], [0:1:0], [0:0:1]\}$, which is trivially colorable.

Case 2: $3 \nmid n, 2 \mid n$

By Lemma ??, no all-nonzero orthogonal pairs exist. All triads have the form $\{v_1, v_2, e_k\}$ where v_1, v_2 are one-zero rays in the same coordinate plane and e_k is the corresponding axis ray.

[Perfect matching] For even n , the orthogonality graph on one-zero rays in each coordinate plane is a perfect matching.

The one-zero rays in the $z = 0$ plane are projectively $(1, \zeta^p, 0)$ for $p \in \{0, \dots, n-1\}$. Two such rays p, q are orthogonal iff $1 + \zeta^{q-p} = 0$, i.e., $q - p \equiv n/2 \pmod{n}$. Each ray has exactly one partner, giving $n/2$ disjoint edges.

[Plane independence] One-zero rays from different coordinate planes are never orthogonal.

$$\langle (\zeta^a, \zeta^b, 0) | (\zeta^c, 0, \zeta^d) \rangle = \zeta^{c-a} \neq 0.$$

An explicit coloring: choose one axis ray as green (say e_1), the other two as red. The e_1 -plane rays are all forced red (orthogonality exclusion). In each of the other two planes, the matched pairs can be independently 2-colored (one green, one red per pair). By Lemma ??, no cross-plane conflicts arise.

Case 3: $3 \mid n, 2 \nmid n$

This is the principal case. Three-term cancellations create triads among all-nonzero rays, but we show these triads are completely isolated.

[Projective collapse] For any all-nonzero ray $v = (\zeta^a, \zeta^b, \zeta^c)$ in S_n with $3 \mid n$, there are exactly two projectively distinct all-nonzero rays orthogonal to v .

Let $\omega = \zeta^{n/3}$. The orthogonality condition $\langle v | w \rangle = 0$ with w all-nonzero requires the coordinate differences $(b_1 - a_1, b_2 - a_2, b_3 - a_3) \pmod{n}$ to be a permutation of $(0, n/3, 2n/3)$ (by Lemma ??). The six permutations split into two projective equivalence classes.

Even permutations $\{(0, n/3, 2n/3), (n/3, 2n/3, 0), (2n/3, 0, n/3)\}$ yield:

$$\begin{aligned} w_+ &= (\zeta^a, \omega \zeta^b, \omega^2 \zeta^c), \\ &= \omega^{-1} \cdot (\omega \zeta^a, \omega^2 \zeta^b, \zeta^c), \\ &= \omega^{-2} \cdot (\omega^2 \zeta^a, \zeta^b, \omega \zeta^c). \end{aligned}$$

All three are projectively equivalent (they differ by the scalar factors $1, \omega^{-1}, \omega^{-2}$).

Odd permutations $\{(0, 2n/3, n/3), (n/3, 0, 2n/3), (2n/3, n/3, 0)\}$ similarly collapse to

$$w_- = (\zeta^a, \omega^2 \zeta^b, \omega \zeta^c).$$

Since $\omega \neq 1$, neither w_+ nor w_- is projectively equivalent to v , and $w_+ \not\sim w_-$ (else $\omega = \omega^2$, forcing $\omega = 1$).

[Triad uniqueness] The triple $\{v, w_+, w_-\}$ is a triad, and it is the unique triad containing v among all-nonzero rays.

$\langle w_+ | w_- \rangle = 1 + \bar{\omega} \cdot \omega^2 + \bar{\omega}^2 \cdot \omega = 1 + \omega + \omega^2 = 0$ (using $\bar{\omega} = \omega^2$). So $\{v, w_+, w_-\}$ is mutually orthogonal. Uniqueness follows from Lemma ??: v has exactly two all-nonzero orthogonal neighbors.

[One-zero isolation] For odd n , no one-zero ray participates in any triad.

A triad containing $(\zeta^a, \zeta^b, 0)$ would require a second ray u with $u_3 = 0$ (forced by orthogonality to the complementary axis ray $(0, 0, \zeta^c)$) satisfying $\langle v | u \rangle = \zeta^{c-a} + \zeta^{c-b} = 0$, i.e., $-1 \in \langle \zeta \rangle$. By Lemma ??, this requires $2 \mid n$.

The triad structure of S_n for odd n with $3 \mid n$ is now determined: one axis triad $\{e_1, e_2, e_3\}$ plus $n^2/3$ all-nonzero triads (since the n^2 all-nonzero rays partition into triples by Lemma ??), with every triad isolated—no two sharing a ray. The coloring problem decomposes into $n^2/3 + 1$ independent single-triad subproblems, each trivially satisfiable by assigning one 1 per triad.

DISCUSSION

The factorization $6 = 2 \times 3$ in Theorem ?? reflects two independent algebraic mechanisms, each necessary for KS-uncolorability:

Phase cancellation ($3 \mid n$): the identity $1 + \omega + \omega^2 = 0$ creates orthogonal triads among all-nonzero rays. Without it, no all-nonzero pair is orthogonal, and triads are restricted to zero-coordinate configurations.

Interlocking ($2 \mid n$): the element $-1 \in \langle \zeta \rangle$ creates two-term cancellations that link triads together. Without it, the projective collapse of Lemma ?? ensures every triad is isolated, making the coloring problem decomposable.

This mirrors the ring-theoretic result of Cortez, Morales, and Reyes [?]: $\mathbb{Z}[1/6]$ is the minimal ring supporting KS contextuality in dimension 3 precisely because both $1/2$ and $1/3$ are needed in the denominators of the projection matrices $P_v = vv^\dagger / \|v\|^2$. Computing their $N(S) = \text{lcm}\{\|v\|^2 : v \in S\}$ invariant for the Eisenstein 33-set gives $N = 6$, confirming it realizes the minimal ring. Theorem ?? shows that this minimality is not an artifact of a particular construction but a structural feature of the cyclotomic family: among all n th-root alphabets, exactly those with $6 \mid n$ inherit both mechanisms.

More broadly, the proof technique—classifying orthogonalities by the length of the underlying vanishing sum,

then analyzing the resulting constraint graph—may extend to other algebraic families of KS constructions. The projective collapse phenomenon (Lemma ??), where group-theoretic symmetry forces orthogonal neighbors to coalesce, suggests a representation-theoretic approach to KS colorability that goes beyond the combinatorial methods currently dominant in the field.

The author acknowledges the use of Claude (Anthropic) for computational assistance and proof verification. SAT solving used PySAT with Glucose4. Code is available at <https://github.com/michaelkernaghan/contextuality>.

-
- [1] S. Kochen and E. P. Specker, *J. Math. Mech.* **17**, 59 (1967).
 - [2] A. Cabello, *Phys. Rev. Lett.* (2025); arXiv:2508.07335.
 - [3] V. Cortez, R. Morales, and E. J. Reyes, arXiv:2211.13216 (2022).
 - [4] J. H. Conway and S. Kochen, reported in A. Peres, *Quantum Theory: Concepts and Methods* (Kluwer, 1993).
 - [5] S. Trandafir and A. Cabello, *Phys. Rev. A* **111**, 052204 (2025); arXiv:2501.11640.
 - [6] Z. Li, C. Bright, and V. Ganesh, in *Proceedings of IJCAI-24*, 2105 (2024); arXiv:2306.13319.